SHAMIM MAHBUB

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Senior Python, Generative AI & Agentic Systems Engineer with 5+ years building production-grade AI solutions. Currently owning the agent runtime, retrieval, and evaluation layers of a consumer conversational assistant. Expert in multi-agent orchestration (LangGraph), RAG and hybrid retrieval, evaluation harnesses, and MCP-based tool integration. Proficient in LLM fine-tuning (LoRA/QLoRA), reasoning model evaluation (o3/DeepSeek-R1), multimodal AI, and LLMOps pipelines. Track record of delivering measurable business impact for clients in the US, Singapore, and Bangladesh.

PROFESSIONAL EXPERIENCE

Senior Python & AI Agent Engineer — HeartStamp, Inc. **Sep 2026 – Present**
Las Vegas, USA (Remote)

- Own the agent runtime of Stampy, HeartStamp's conversational greeting-card assistant, built on LangGraph and LangChain; move recorded decisions from model output into deterministic, validated system state with checkpointed, resumable conversations and Pydantic-validated tool schemas.
- Own hybrid retrieval over the card catalogue on PostgreSQL + pgvector (HNSW), combining full-text and vector ranking with two-stage reranking; measure ranking quality on a labeled query set with Recall@k, MRR, nDCG, and precision@k.
- Build the evaluation harness that runs in CI on every pull request, asserting structured-output validity, grounding, tool allowlists, and idempotency, with LangSmith tracing and production metrics for agent quality and retrieval drift.

Senior AI Engineer — Omnexis AI **Feb 2026 – Sep 2026**
Remote

- Led design and development of memory-augmented, multi-agent orchestration systems, integrating long-term and episodic memory stores to improve task continuity across multi-turn agent workflows.
- Architected RAG and agentic RAG pipelines with hybrid retrieval (vector + keyword) to improve grounding accuracy and reduce hallucination rates in production LLM applications.
- Owned end-to-end LLMOps for deployed agents: prompt versioning, regression testing, drift monitoring, and guardrail enforcement against OWASP LLM Top 10 risks.
- Collaborated cross-functionally with product and backend teams to integrate LLM/agent services into existing Python/API-based systems, ensuring scalable, low-latency inference in production.

GenAI Engineer — Artech Digital **Nov 2024 – Jan 2026**
San Francisco, USA (Remote)

- Led end-to-end development of agentic AI systems for enterprise clients using LangGraph, CrewAI, and MCP-based tool integration, enabling autonomous multi-step task execution across complex workflows.
- Designed and deployed a WhatsApp-based e-commerce platform powered by LangGraph multi-agent orchestration; integrated M-Pesa Daraja APIs for fully automated end-to-end transactions.
- Built an Enterprise Bid Generation Agent (LangGraph + RAG) that auto-generates RFP responses by analyzing requirements and retrieving past proposals, reducing bid preparation time by 65%.
- Evaluated and benchmarked reasoning models (OpenAI o3, DeepSeek-R1) for client use cases, advising on model selection trade-offs between latency, cost, and accuracy.

Machine Learning Engineer — Spekter GmbH **Jan 2024 – Nov 2024**
Adabor, Dhaka, Bangladesh

- Developed anomaly detection models for industrial sensor data using time-series deep learning (LSTM, Transformer-based architectures), reducing false-positive alerts by ~30%.
- Implemented AWS CloudWatch monitoring pipelines and automated alerting for ML model performance degradation in production.

Software Engineer – AI & NLP — SkyRanko **Jan 2023 – Dec 2023**
Rajshahi, Bangladesh

- Built NLP pipeline for paraphrasing and summarization of English text using fine-tuned T5/BART models.
- Developed automated web scraping systems for data collection pipelines feeding downstream ML models.

Software Engineer – NLP — Zibew E-Commerce PTE Ltd. **Mar 2022 – Dec 2022**
Singapore (Remote)

- Developed and deployed AI chatbots for multiple client platforms; researched LLM-based conversational AI to replace rule-based systems.

Artificial Intelligence Engineer — Opus Technology Ltd. **Jul 2021 – Feb 2022**
Dhaka, Bangladesh

- Built a Reinforcement Learning bot for CS:GO using PPO/DQN; developed company website chatbot using intent classification.

TECHNICAL SKILLS

Languages: Python (Expert), SQL, C (Intermediate)

Agentic & GenAI: LLM Fine-tuning, Prompt Engineering, RAG, Agentic RAG, MCP (Model Context Protocol), AI Agents, Multimodal AI, Reasoning Models (o3/o4, DeepSeek-R1, Gemini Thinking), LLMOps, Evaluation Harnesses, Guardrails / Red Teaming

Frameworks: LangChain, LangGraph, CrewAI, LlamaIndex, AutoGen (AG2), FastMCP, Pydantic, Keras, TensorFlow, PyTorch

Libraries: Transformers, FAISS, Sentence-Transformers, NLTK, SpaCy, NumPy, Pandas, OpenCV, Matplotlib, SciPy

Vector DBs & Retrieval: PostgreSQL + pgvector (HNSW), Pinecone, Chroma, Qdrant, Weaviate; hybrid ranking, reranking, Recall@k / MRR / nDCG

Cloud & MLOps: AWS (SageMaker, Lambda, S3, ECS), Azure (OpenAI Service, ML Workspace), Vertex AI, MLflow, Weights & Biases, LangSmith, Docker, FastAPI, Flask, GitHub Actions

Tools & Platforms: HuggingFace, ComfyUI, Gradio, Streamlit, Selenium, Git & GitHub, MySQL, SQLServer, Redis, Celery, Prometheus, Grafana, Linux

SELECTED PROJECTS

MCP-Integrated Enterprise RAG System — Built an MCP server (FastMCP) wrapping a domain-specific RAG pipeline, enabling portable tool integration across Claude, VS Code, and custom agent backends. Improved factual accuracy 47% vs. base LLM.

Reasoning Model Evaluation Framework — Designed a benchmarking system comparing o3, DeepSeek-R1, and Gemini Thinking models across enterprise use cases; delivered actionable model-selection guidance reducing inference costs by 30%.

Enterprise Bid Generation Agent — LangGraph multi-agent system that auto-generates customized RFP responses by analyzing requirement docs and retrieving past proposals; reduced bid preparation time by 65%.

GEO Intelligence Platform — Scalable LLM evaluation platform generating prompt variants, running multi-temperature sampling experiments, and analyzing AI brand visibility across generative search engines with cost tracking.

Medical Knowledge Assistant — Healthcare-focused RAG system using specialized medical embeddings and clinical knowledge bases; achieved 93% accuracy on physician-verified test cases.

LLM Fine-tuning Pipeline (LoRA/QLoRA) — Automated pipeline for parameter-efficient fine-tuning of open-source LLMs on proprietary data; reduced training costs by 68% vs. full fine-tuning.

WhatsApp Agentic Commerce Platform — LangGraph-powered conversational commerce on WhatsApp with AI-driven trade-in valuation, product recommendations, and integrated M-Pesa STK push payments — fully automated end-to-end.

PUBLICATIONS

- Hasan, M. J., Mahbub, S., et al. "Rice disease identification and classification by integrating SVM with deep CNN." ICASERT 2019, IEEE. (141 citations)
- Wahid, M. F., Hasan, M. J., Alom, M. S., & Mahbub, S. "Performance Analysis of ML Techniques for Microscopic Bacteria Image Classification." ICCCNT 2019, IEEE. (15 citations)
- Mahbub, S., et al. "Comparison of Handcrafted and Deep Neural Network Feature Extraction for Recognizing Bangla Sign Digits." ICCCNT 2021, IEEE.

EDUCATION

B.Sc. (Engg.) in Electrical & Electronic Engineering — Hajee Mohammad Danesh Science & Technology University (HSTU) **2016 – 2019**

Dinajpur, Bangladesh

- CGPA: 3.34 / 4.00 | Research: Sleep Apnea detection via ECG (Deep Learning), Retinal Exudate Classification (DCNN)

RELEVANT CERTIFICATIONS & COURSEWORK

Advanced Generative AI Development • Building Production LLM Applications • RAG System Design & Implementation • LLM Fine-tuning & Model Alignment • LangChain & LlamaIndex for AI Applications • AI Agent Design & Orchestration • Vector Databases & Semantic Search • Multimodal AI Systems Development • Deep Learning Specialization • NLP Specialization • ChatGPT Prompt Engineering for Developers

LANGUAGES & LEADERSHIP

Languages: Bangla (Native), English (Professional)

Leadership: IEEE HSTU Student Branch – Membership Committee Convener; Joint Secretary, EEE Club of HSTU; Volunteer, National High School Programming Contest